LAB 07— Probability & Statistics for ML (Naive Bayes)

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1. Objective

Build a complete system that:

- Understands statistics (mean, variance, probability)
- Applies them on a dataset
- Uses them in a Machine Learning model
- Deploys as a GUI application

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2. Part A — Core Statistical Concepts (Foundation)

2.1 Mean (Average)

$$\text{Mean} = \frac{\sum x_i}{n}$$

- Adds all values and divides by total count
- Represents central value

2.2 Median

- Middle value after sorting
- Handles outliers better than mean

2.3 Variance

$$\text{Variance} = \frac{\sum (x_i - \mu)^2}{n}$$

- Measures spread of data
- Uses squared differences

2.4 Standard Deviation

$$\sigma = \sqrt{\text{Variance}}$$

- Indicates how far values deviate from mean

2.5 Probability

$$P(A) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

Example:

$$P(\text{Pass}) = \frac{7}{10} = 0.7$$

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3. Part B — Dataset & Statistical Analysis

3.1 Dataset (Real CSV)

Columns:

- Study_Hours
- Attendance
- Assignment
- Marks

3.2 Target Variable Creation

Rule:

- Pass = Marks ≥ 60
- Fail = Marks < 60

3.3 Statistical Calculations

Mean

$$\text{Mean Marks} = \frac{\sum Marks}{n}$$

Variance & Standard Deviation

- Measure distribution spread

Probability

$$P(\text{Pass}) = \frac{\text{Pass Students}}{\text{Total Students}}$$

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4. Part C — Machine Learning Model (Complete Explanation)

4.1 What is Naive Bayes?

Naive Bayes is a **probabilistic classification algorithm** based on Bayes' Theorem.

It predicts class (Pass/Fail) using probability.

4.2 Bayes Theorem

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

$$P(A)$$

$$P(B \mid A)$$

$$P(B \mid \neg A)$$

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \approx 0.68,\ P(B) \approx 0.25$$

Posterior = useful evidence / total evidence

P(B)=0.25

P(B|A)P(A)=0.17P(A|B)~0.68

4.3 Meaning of Each Term

- $P(A \mid B)$: Probability of class given data (**Posterior**)

- $P(B \mid A)$: Probability of data given class (**Likelihood**)
- $P(A)$: Prior probability of class
- $P(B)$: Total probability of data

4.4 Step-by-Step Working in This Project

We want:

$$P(\text{Pass} \mid \text{Study Hours, Attendance, Assignment})$$

Step 1 — Calculate Prior

$$P(\text{Pass}) = \frac{\text{Pass Students}}{\text{Total}}$$
$$P(\text{Fail}) = \frac{\text{Fail Students}}{\text{Total}}$$

Step 2 — Calculate Likelihood

For each feature:

- Probability of Study Hours given Pass
- Probability of Attendance given Pass
- Probability of Assignment given Pass

Naive assumption:

Features are independent

So:

$$P(X \mid Pass) = P(x_1 \mid Pass) \cdot P(x_2 \mid Pass) \cdot P(x_3 \mid Pass)$$

Step 3 — Posterior Calculation

$$P(Pass \mid X) = P(X \mid Pass) \cdot P(Pass)$$
$$P(Fail \mid X) = P(X \mid Fail) \cdot P(Fail)$$

Step 4 — Final Decision

- If $P(Pass \mid X) > P(Fail \mid X)$→ PASS
- Else → FAIL

4.5 Why Gaussian Naive Bayes?

Because features are **continuous numbers**.

It assumes data follows **normal distribution**:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

4.6 Important Assumption

Naive Bayes assumes:

All features are independent

In real life this is not always true
But still works well in practice

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5. Part D — Complete Implementation (Code)

```
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

st.title("Student Performance Predictor")

uploaded_file = st.file_uploader("Upload CSV", type=["csv"])

if uploaded_file:
    df = pd.read_csv(uploaded_file)
```

```
# Create target
df["Result"] = df["Marks"].apply(lambda x: 1 if x >= 60 else 0)

st.write(df.head())

# Statistics
st.subheader("Statistical Report")

st.write("Mean:", df["Marks"].mean())
st.write("Median:", df["Marks"].median())
st.write("Variance:", df["Marks"].var())
st.write("Std Dev:", df["Marks"].std())

# Probability
pass_prob = len(df[df["Result"] == 1]) / len(df)
st.write("Probability of Passing:", pass_prob)

# Visualization
fig, ax = plt.subplots()
ax.hist(df["Marks"])
st.pyplot(fig)

# ML Model
X = df[["Study_Hours", "Attendance", "Assignment"]]
y = df["Result"]

model = GaussianNB()
model.fit(X, y)

pred = model.predict(X)
acc = accuracy_score(y, pred)

st.write("Model Accuracy:", acc)

# Prediction UI
st.subheader("Predict")

study = st.slider("Study Hours", 0, 10)
att = st.slider("Attendance", 0, 100)
```

```
assign = st.slider("Assignment", 0, 100)

if st.button("Predict"):
    result = model.predict([[study, att, assign]])

    if result[0] == 1:
        st.success("PASS")
    else:
        st.error("FAIL")
```

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6. Part E — Output Explanation

System Produces

1. Statistical Report

- Mean
- Median
- Variance
- Standard deviation

2. Probability

- Passing probability

3. ML Model

- Trained Naive Bayes
- Predictions

4. Accuracy

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

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7. Part F — Interpretation

- Higher study hours → higher chance of passing
- Model uses probability, not rules

- Predictions are based on learned patterns